



Sol final ML 2021-2022 - Solution of Machine learning exam WS21

Machine learning (Technische Universität München)



Scanne, um auf Studocu zu öffnen

Ecorrection

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Maschinelles Lernen

Exam: IN2064 / Endterm

Date: Thursday 17th February, 2022

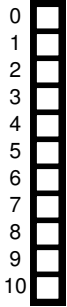
Examiner: Prof. Dr. Stephan Günnemann

Time: 17:00 – 19:00

Working instructions

- You have to sign the code of conduct. (Typing your name is fine).
- You have to either print this document, solve the problems on paper and then scan your solutions OR paste scans/pictures of your handwritten, on-paper solutions into the solution boxes in this PDF.
- You must not use any other means of creating a submission (e.g. digital pen on a tablet).
- Make sure that the **QR codes are visible** on every uploaded page. Otherwise, we cannot grade your submission.
- **You must solve the specified version of the problem.** Different problems may have different versions: e.g. Problem 1 (Version A), Problem 5 (Version C), etc. If you solve the wrong version you get **zero** points.
- Only write on the provided sheets, **submitting your own additional sheets is not possible**.
- The last pages (after problem 11) can be used as scratch paper.
- All sheets (save for empty scratch paper) have to be submitted to the upload queue. Missing pages will be considered empty.
- **Only use a black or blue color (no red or green)! Pencils are allowed.**
- **For problems that say "Justify your answer" you only get points if you provide a valid explanation.**
- **For problems that say "Derive" you only get points if you provide a valid mathematical derivation.**
- **For problems that say "Prove" you only get points if you provide a valid mathematical proof.**
- If a problem does not say "Justify your answer", "Derive" or "Prove" it's sufficient to only provide the correct answer.
- Instructor announcements and clarifications will be posted **on Piazza** with email notifications.
- Exercise duration - 120 minutes.

Problem 1 (All versions) (10 credits)



The posterior distribution is proportional to:

$$\begin{aligned}\mathbb{P}(\{x_1, \dots, x_N\} \mid \theta) \times \mathbb{P}(\theta \mid \lambda, \alpha) &= \prod_i \mathbb{P}(x_i \mid \theta) \times \mathbb{P}(\theta \mid \lambda, \alpha) \\ &= \frac{1}{\theta^N} \mathbf{1}_{\max(x_i) \leq \theta} \frac{1}{\theta^{\alpha+1}} \mathbf{1}_{\max(\lambda) \leq \theta} \alpha \lambda^\alpha \\ &\propto \frac{1}{\theta^{N+\alpha+1}} \mathbf{1}_{\max(x_1, \dots, x_N, \lambda) \leq \theta}\end{aligned}$$

Writing posterior = likelihood $\times$ prior ✓ ✓ .

Writing likelihood as a product of x_i observations ✓ ✓ .

Writing formula for likelihood and prior based with indicator functions ✓ ✓ .

Simplify formula for posterior correctly ✓ ✓ .

Recognize the shape of the prior distribution. You should not remove points for the name "Pareto" ✓ ✓ .

Derive keyword, i.e. 0 points for only stating the solution.

We recognize that the posterior distribution is $\mathbb{P}(\theta \mid \{x_1, \dots, x_N\}, \lambda, \alpha) = \mathbb{P}(\theta \mid \lambda_{\text{new}}, \alpha_{\text{new}}) = \text{Pareto}(\lambda_{\text{new}} = \max(x_1, \dots, x_N, \lambda), \alpha_{\text{new}} = N + \alpha)$.

Problem 2 (All versions) (8 credits)

☐	0
☐	1
☐	2
☐	3
☐	4
☐	5
☐	6
☐	7
☐	8

We write the optimal solution of the new problem:

$$\mathbf{w}_{new}^* = (\mathbf{X}_{new}^T \mathbf{X}_{new})^{-1} \mathbf{X}_{new}^T \frac{1}{\sigma} \mathbf{y}$$

We want to have:

$$(\mathbf{X}_{new}^T \mathbf{X}_{new})^{-1} \mathbf{X}_{new}^T \frac{1}{\sigma} \mathbf{y} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

This is achieved if $\mathbf{X}_{new} = \frac{1}{\sigma} \mathbf{X}$

Write correctly the closed form solution of the new problem ✓✓.

Specify the equality between the solution of the previous problem and the new problem ✓✓.

Write the correct solution ✓✓✓✓.

Justify keyword, i.e. 0 points for only stating the solution.

Sample Solution

Correction Notes

Problem 3 (Version A) (3 credits)

0
1

a)

Distance	Assignment (a-c)
L_2 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _2 = \sqrt{\sum_i (\mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)})^2}$	b
L_1 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _1 = \sum_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	a
L_∞ -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _\infty = \max_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	c

Iff all three are correct ✓.

0
1

b)

Class 2 (triangle) ✓

0
1

c)

Class 1 (x) ✓

Problem 3 (Version B) (3 credits)

a)

0
1

Distance	Assignment (a-c)
L_2 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _2 = \sqrt{\sum_i (\mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)})^2}$	b
L_∞ -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _\infty = \max_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	c
L_1 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _1 = \sum_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	a

Iff all three are correct ✓.

b)

0
1

Class 2 (x) ✓

c)

0
1

Class 1 (triangle) ✓

Problem 3 (Version C) (3 credits)

0
1

a)

Distance	Assignment (a-c)
L_∞ -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _\infty = \max_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	c
L_2 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _2 = \sqrt{\sum_i (\mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)})^2}$	b
L_1 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _1 = \sum_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	a

Iff all three are correct ✓.

0
1

b)

Class 2 (triangle) ✓

0
1

c)

Class 1 (x) ✓

Problem 3 (Version D) (3 credits)

a)

0
1

Distance	Assignment (a-c)
L_∞ -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _\infty = \max_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	c
L_1 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _1 = \sum_i \mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)} $	a
L_2 -distance: $\ \mathbf{x}^{(1)} - \mathbf{x}^{(2)}\ _2 = \sqrt{\sum_i (\mathbf{x}_i^{(1)} - \mathbf{x}_i^{(2)})^2}$	b

Iff all three are correct ✓.

b)

0
1

Class 2 (x) ✓

c)

0
1

Class 1 (triangle) ✓

Problem 4 (Version A) (6 credits)

0 ☐
1 ☐
2 ☐

a)

1. Two isotropic bivariate Gaussian distributions with identical covariance matrices
2. Inconclusive, since both models' assumptions match the data.

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0 ☐
1 ☐
2 ☐

b)

1. Identical covariance matrices but correlated features
2. Linear Discriminant Analysis since the covariance matrices match for both classes and the features are correlated (alternative: Naïve Bayes' conditional independence assumption is clearly violated).

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0 ☐
1 ☐
2 ☐

c)

1. Covariance matrices do not match and no apparent correlation between features
2. Naïve Bayes' since the features are conditionally independent and variances mismatch between classes $\Rightarrow$ quadratic decision boundary (alternative: Linear Discriminant Analysis' assumption of equal covariance matrices is clearly violated).

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

Problem 4 (Version B) (6 credits)

a)

1. Identical covariance matrices but correlated features
2. Linear Discriminant Analysis since the covariance matrices match for both classes and the features are correlated (alternative: Naïve Bayes' conditional independence assumption is clearly violated).

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

b)

1. Two isotropic bivariate Gaussian distributions with identical covariance matrices
2. Inconclusive, since both models' assumptions match the data.

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

c)

1. Covariance matrices do not match and no apparent correlation between features
2. Naïve Bayes' since the features are conditionally independent and variances mismatch between classes $\Rightarrow$ quadratic decision boundary (alternative: Linear Discriminant Analysis' assumption of equal covariance matrices is clearly violated).

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

Problem 4 (Version C) (6 credits)

0 ☐
1 ☐
2 ☐

a)

1. Covariance matrices do not match and no apparent correlation between features
2. Naïve Bayes' since the features are conditionally independent and variances mismatch between classes $\Rightarrow$ quadratic decision boundary (alternative: Linear Discriminant Analysis' assumption of equal covariance matrices is clearly violated).

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0 ☐
1 ☐
2 ☐

b)

1. Identical covariance matrices but correlated features
2. Linear Discriminant Analysis since the covariance matrices match for both classes and the features are correlated (alternative: Naïve Bayes' conditional independence assumption is clearly violated).

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0 ☐
1 ☐
2 ☐

c)

1. Two isotropic bivariate Gaussian distributions with identical covariance matrices
2. Inconclusive, since both models' assumptions match the data.

If correct assignment and reasonable explanation ✓ ✓
Otherwise 0 (i.e. even for correct assignment without explanation)

Problem 4 (Version D) (6 credits)

a)

1. Covariance matrices do not match and no apparent correlation between features
2. Naïve Bayes' since the features are conditionally independent and variances mismatch between classes $\Rightarrow$ quadratic decision boundary (alternative: Linear Discriminant Analysis' assumption of equal covariance matrices is clearly violated).

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

b)

1. Two isotropic bivariate Gaussian distributions with identical covariance matrices
2. Inconclusive, since both models' assumptions match the data.

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

c)

1. Identical covariance matrices but correlated features
2. Linear Discriminant Analysis since the covariance matrices match for both classes and the features are correlated (alternative: Naïve Bayes' conditional independence assumption is clearly violated).

If correct assignment and reasonable explanation ✓✓
Otherwise 0 (i.e. even for correct assignment without explanation)

0
1
2

Problem 5 (All versions) (10 credits)

0
1
2
3
4

a)

Prove keyword, i.e. 0 points for only stating the solution.

From the lecture we know: Let $f_1 : \mathbb{R}^d \rightarrow \mathbb{R}$ and $f_2 : \mathbb{R}^d \rightarrow \mathbb{R}$ be convex functions, and $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a concave function, then: (1) $h(\mathbf{x}) = f_1(\mathbf{x}) + f_2(\mathbf{x})$ is convex, (2) $h(\mathbf{x}) = \max \{f_1(\mathbf{x}), f_2(\mathbf{x})\}$ is convex, (3) $h(\mathbf{x}) = c \cdot f_1(\mathbf{x})$ is convex if $c \geq 0$, (4) $h(\mathbf{x}) = c \cdot g(\mathbf{x})$ is convex if $c \leq 0$, (5) $h(\mathbf{x}) = f_1(\mathbf{A}\mathbf{x} + \mathbf{b})$ is convex ($\mathbf{A}$ matrix, $\mathbf{b}$ vector), and (6) $h(\mathbf{x}) = m(f_1(\mathbf{x}))$ is convex if $m : \mathbb{R} \rightarrow \mathbb{R}$ is convex and nondecreasing.

Consider the function $e_i(\mathbf{x}) = x_i$ which is clearly convex in $\mathbf{x}$ since it is constant in all dimensions but the i -th, in which it is linear. From the definition of convexity, it is easy to see that $e_i(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) \leq \lambda e_i(\mathbf{x}) + (1 - \lambda)e_i(\mathbf{y})$ holds. (✓)

(By induction, rule (2) also holds for more than two arguments.) Thus, by rule (2) $\max_{i=1, \dots, n} x_i$ is convex (plugging in $e_i(\mathbf{x})$ for $f_i(\mathbf{x})$). (✓)

Equivalently $\min_{i=1, \dots, n} x_i$ is concave. By rule (4) it follows that $-\min_{i=1, \dots, n} x_i$ is convex. (✓)

Last, by rule (1) it follows that $f(\mathbf{x})$ is convex in $\mathbf{x}$. (✓)

0
1
2
3
4
5
6

b)

Prove keyword, i.e. 0 points for only stating the solution.

Solution with the wrong hint (as stated on sheet) $\text{median}(\mathbf{x}) = \min_{t \in \mathbb{R}} \|\mathbf{x} - t\mathbf{1}\|_1$

This can be solved easiest with a counter-example. Assume the instances $\mathbf{x}_1 = [0, 1, 1]^T$ and $\mathbf{x}_2 = [2, 1, 1]^T$ as well as $\lambda = 0.5$ (i.e. $\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2 = [1, 1, 1]^T$). Hence, in all cases the median is $t^* = 1$, i.e. $t^* = \arg \min_{t \in \mathbb{R}} \|\mathbf{x} - t\mathbf{1}\|_1$. (✓ ✓ ✓)

$\Rightarrow \text{median}(\mathbf{x}_1) = 1, \text{median}(\mathbf{x}_2) = 1, \text{and } \text{median}(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) = 0$ (✓)

$\Rightarrow g(\mathbf{x}_1) = 1, g(\mathbf{x}_2) = 1, \text{and } g(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) = 3$ (✓)

$\Rightarrow g(\lambda\mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2) > \lambda g(\mathbf{x}_1) + (1 - \lambda)g(\mathbf{x}_2)$ (✓)

Therefore, $g(\mathbf{x})$ is not convex in $\mathbf{x}$.

Solution with the corrected hint (as updated on Piazza) $\text{median}(\mathbf{x}) = \arg \min_{t \in \mathbb{R}} \|\mathbf{x} - t\mathbf{1}\|_1$

Option 1: direct application of the convexity definition

For two arbitrary $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, by definition of $g(\mathbf{x})$: $t_x^* = \text{median}(\mathbf{x}) = \arg \min_t \|\mathbf{x} - t \cdot \mathbf{1}\|_1$ and $t_y^* = \text{median}(\mathbf{y}) = \arg \min_t \|\mathbf{y} - t \cdot \mathbf{1}\|_1$. Moreover, $t_\lambda^* = \text{median}(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) = \arg \min_t \|\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} - t \cdot \mathbf{1}\|_1$ for an arbitrary $\lambda \in [0, 1]$. Also note that $\sum_{i=1}^n |x_i - t| = \|\mathbf{x} - t\mathbf{1}\|_1$.

$$\begin{aligned} \lambda g(\mathbf{x}) + (1 - \lambda)g(\mathbf{y}) &= \lambda \frac{1}{n} \|\mathbf{x} - t_x^* \mathbf{1}\|_1 + (1 - \lambda) \frac{1}{n} \|\mathbf{y} - t_y^* \mathbf{1}\|_1 \\ &= \frac{1}{n} [\|\lambda\mathbf{x} - \lambda t_x^* \mathbf{1}\|_1 + \|(1 - \lambda)\mathbf{y} - (1 - \lambda)t_y^* \mathbf{1}\|_1] \quad (\checkmark) \\ &\stackrel{(1)}{\geq} \frac{1}{n} \|\lambda\mathbf{x} - \lambda t_x^* \mathbf{1} + (1 - \lambda)\mathbf{y} - (1 - \lambda)t_y^* \mathbf{1}\|_1 \quad (\checkmark \checkmark) \\ &\stackrel{(2)}{\geq} \frac{1}{n} \|\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} - t_\lambda^* \mathbf{1}\|_1 \quad (\checkmark \checkmark) \\ &= g(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y}) \quad (\checkmark) \end{aligned}$$

(1) follows from triangle inequality ($\|\mathbf{a}\| + \|\mathbf{b}\| \geq \|\mathbf{a} + \mathbf{b}\|$) and (2) from the definition of t_λ^* above. Since $\lambda g(\mathbf{x}) + (1 - \lambda)g(\mathbf{y}) \geq g(\lambda\mathbf{x} + (1 - \lambda)\mathbf{y})$ it follows that $g(\mathbf{x})$ is convex.

Option 2: argument via convexity preserving operations (as much as possible)

An integral part is to analyze the convexity of $\sum_{i=1}^n |x_i - t| = \|\mathbf{x} - t\mathbf{1}\|_1$. First, observe that $\mathbf{x} - t\mathbf{1}$ is a linear function in $\mathbf{x}$ (as well as t). Equivalently, we can write $\mathbf{1}\mathbf{x} - t\mathbf{1} = \mathbf{A}\mathbf{x} + \mathbf{b}$ (✓). Moreover, $\|\mathbf{x}\|_1$ is convex but *not nondecreasing* for $\mathbf{x} \in \mathbb{R}^n$ (i.e. rule 6 does not apply). From rule 5 it follows that $\|\mathbf{x} - t\mathbf{1}\|_1$ is convex in $\mathbf{x}$ (as well as t). (✓✓)

By rule (3) $g(\mathbf{x})$ is convex if $\sum_{i=1}^n |x_i - \text{median}(\mathbf{x})|$ is convex since $\frac{1}{n} > 0$. (✓)

$\text{median}(\mathbf{x}) = \arg \min_{t \in \mathbb{R}} \|\mathbf{x} - t\mathbf{1}\|_1$. Hence, we can write $g(\mathbf{x}) = \frac{1}{n} \|\mathbf{x} - \text{median}(\mathbf{x})\mathbf{1}\|_1 = \frac{1}{n} \min_{t \in \mathbb{R}} \|\mathbf{x} - t\mathbf{1}\|_1$. (✓)

Now it is left to proof that $g(\mathbf{x}) = \min_{t \in \mathbb{R}} f(\mathbf{x}, t)$ is convex for a function $f(\mathbf{x}, t)$ that convex in both $\mathbf{x}$ and t (✓). For arbitrary $\mathbf{x}_1$ and $\mathbf{x}_2$ we define $t_1 = \arg \min_{t \in \mathbb{R}} f(\mathbf{x}_1, t)$ and $t_2 = \arg \min_{t \in \mathbb{R}} f(\mathbf{x}_2, t)$. For an arbitrary $\lambda \in [0, 1]$:

$$\begin{aligned} g(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2) &= \min_{t \in \mathbb{R}} f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2, t) \\ &\leq f(\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2, \lambda t_1 + (1 - \lambda) t_2) \\ &\leq \lambda f(\mathbf{x}_1, t_1) + (1 - \lambda) f(\mathbf{x}_2, t_2) \\ &= \lambda g(\mathbf{x}_1) + (1 - \lambda) g(\mathbf{x}_2) \end{aligned}$$

Hence, $g(\mathbf{x})$ is a convex function.

Problem 6 (Version A) (8 credits)

0 ☐
1 ☐
2 ☐
3 ☐
4 ☐

a)

```
out = x * np.sin(y) ✓✓✓✓  
Also correct: out = x * sin(y)
```

0 ☐
1 ☐
2 ☐
3 ☐
4 ☐

b)

```
N = len(z)  
d_z = np.ones_like(z) / N * d_out ✓✓✓✓  
Also correct: 1 / N * d_out with np.repeat or similar to make vector  
deduct 1 point for missing N  
deduct 1 point for missing d_out  
deduct 1 point for missing np.ones_like(z) or not demonstrating that the output is a vector with  
correct shape
```

Problem 6 (Version B) (8 credits)

a)

`out = np.exp(x) / np.exp(y)` ✓✓✓✓
Also correct: `out = np.exp(x - y)` and without np, or writing e^{x-y}

0
1
2
3
4

b)

`d_z = np.ones_like(z) * d_out` ✓✓✓✓
deduct 2 points for missing `d_out`
deduct 1 point for missing `np.ones_like(z)` or not demonstrating that the output is a vector with correct shape

0
1
2
3
4

Problem 6 (Version C) (8 credits)

0 ☐
1 ☐
2 ☐
3 ☐
4 ☐

a)

`out = np.sin(x * y)` ✓✓✓✓
Also correct without numpy, writing only `sin(x * y)`

0 ☐
1 ☐
2 ☐
3 ☐
4 ☐

b)

`d_z = np.prod(z) / z * d_out` ✓✓✓✓
Also correct: any code that computes the product $\prod_{i \neq j} z_i$
Deduct 1 point if incorrect shape
Deduct 1 point if product of all z without dividing by the current one
Deduct 1 point if missing `d_out`

Problem 6 (Version D) (8 credits)

a)

out = x + x * y ✓✓✓✓

0
1
2
3
4

b)

d_z = 2 * z * d_out ✓✓✓✓
Deduct 2 points if missing z
Deduct 1 point if missing 2
Deduct 1 point if missing d_out

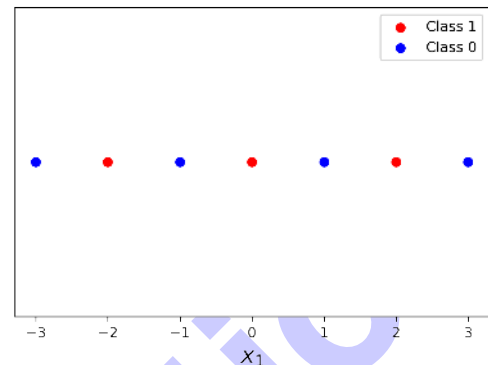
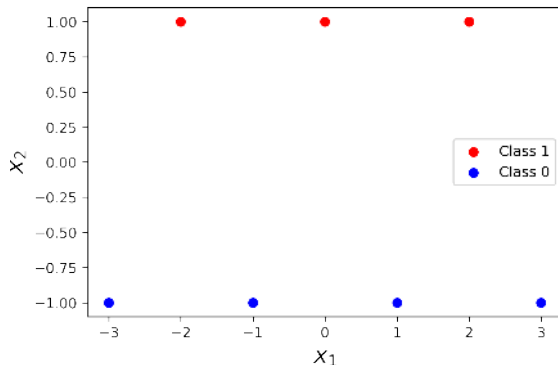
0
1
2
3
4

Problem 7 (All versions) (12 credits)

0	
1	
2	
3	
4	
5	
6	

a)

Consider the 2-dimensional dataset in the left figure, which is clearly linearly separable.



The features are already uncorrelated, meaning PCA will not change the dataset. When we remove the lower-variance dimension x_2 , the data is clearly no longer linearly separable, as shown in the right figure.

6× ✓ for any valid counterexample (an explanation for why it is a valid counterexample is not necessary.)

4× ✓ for a correct approach with some minor mistake.

2× ✓ for providing an intuitive explanation.

0 credits for stating “no” without providing a counterexample.

0	
1	
2	
3	
4	
5	
6	

b)

Proof by contradiction.

Assume that $(\mathbf{X}, \mathbf{y})$ was not linearly separable, but $(\tilde{\mathbf{X}}, \mathbf{y})$ was linearly separable.

Let $\Gamma \in \mathbb{R}^{D \times K}$ be the top- K eigenvectors of the covariance matrix, i.e. the matrix we use to perform dimensionality reduction.

Since $(\tilde{\mathbf{X}}, \mathbf{y})$ is linearly separable, there is a linear classifier $I[\mathbf{w}^T \mathbf{x} \geq b]$ with parameters $\mathbf{w} \in \mathbb{R}^k$ and $b \in \mathbb{R}$ that correctly classifies all points in $(\tilde{\mathbf{X}}, \mathbf{y})$.

Based on the definition of $\tilde{\mathbf{X}}$, i.e. $\tilde{\mathbf{X}} = \Gamma^T \mathbf{X}$, this means that any point in $(\mathbf{X})$ can be correctly classified by first performing dimensionality reduction with PCA and then applying the linear classifier specified above.

However, this procedure in itself is a linear classifier $I[(\mathbf{w}^T \Gamma^T) \mathbf{x} \geq b]$ with weight vector $\mathbf{w}' = \Gamma \cdot \mathbf{w}$.

This contradicts our assumption that $(\mathbf{X}, \mathbf{y})$ is not linearly separable.

6× ✓ for any valid proof.

4× ✓ for a proof with some mistake / partially missing information.

2× ✓ for an informal proof.

Prove keyword, i.e. 0 points for only stating the “yes”.

Problem 8 (Version A) (6 credits)

☐	0
☐	1
☐	2
☐	3
☐	4
☐	5
☐	6

- Loss curve A corresponds to model 1: Both the training and test losses are high — the model *underfits*. This may happen if k is too low.
- Loss curve C corresponds to model 3: The train loss is low, but the test loss increases sharply after a few iterations — this is a clear example of *overfitting*. This may happen if k is too high.
- Loss curve B corresponds to model 2: by exclusion.

- 6× ✓ if all assignments and justifications are correct.
- 4× ✓ if all assignments are correct, but 1/3 justifications is wrong.
- 1 ✓ is subtracted for each completely incorrect statement (e.g., "overfitting the test set"), even if the rest of the answer is correct.

Justify keyword, i.e. 0 points for only stating the solution.

Sample Solution

Correction Notes

Problem 8 (Version B) (6 credits)

0	
1	
2	
3	
4	
5	
6	

- Loss curve B corresponds to model 1: Both the training and test losses are high — the model *underfits*. This may happen if k is too low.
- Loss curve C corresponds to model 3: The train loss is low, but the test loss increases sharply after a few iterations — this is a clear example of *overfitting*. This may happen if k is too high.
- Loss curve A corresponds to model 2: by exclusion.

- 6× ✓ if all assignments and justifications are correct.
- 4× ✓ if all assignments are correct, but 1/3 justifications is wrong.
- 1 ✓ is subtracted for each completely incorrect statement (e.g., "overfitting the test set"), even if the rest of the answer is correct.

Justify keyword, i.e. 0 points for only stating the solution.

Sample Solution

Correction Notes

Problem 8 (Version C) (6 credits)

0
1
2
3
4
5
6

- Loss curve B corresponds to model 1: Both the training and test losses are high — the model *underfits*. This may happen if k is too low.
- Loss curve A corresponds to model 3: The train loss is low, but the test loss increases sharply after a few iterations — this is a clear example of *overfitting*. This may happen if k is too high.
- Loss curve C corresponds to model 2: by exclusion.

- 6× ✓ if all assignments and justifications are correct.
- 4× ✓ if all assignments are correct, but 1/3 justifications is wrong.
- 1 ✓ is subtracted for each completely incorrect statement (e.g., "overfitting the test set"), even if the rest of the answer is correct.

Justify keyword, i.e. 0 points for only stating the solution.

Sample Solution

Correction Notes

Problem 8 (Version D) (6 credits)

0	
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- Loss curve C corresponds to model 1: Both the training and test losses are high — the model *underfits*. This may happen if k is too low.
- Loss curve B corresponds to model 3: The train loss is low, but the test loss increases sharply after a few iterations — this is a clear example of *overfitting*. This may happen if k is too high.
- Loss curve A corresponds to model 2: by exclusion.

- 6× ✓ if all assignments and justifications are correct.
- 4× ✓ if all assignments are correct, but 1/3 justifications is wrong.
- 1 ✓ is subtracted for each completely incorrect statement (e.g., "overfitting the test set"), even if the rest of the answer is correct.

Justify keyword, i.e. 0 points for only stating the solution.

Sample Solution

Correction Notes

Problem 9(All versions) (12 credits)

0
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10
11
12

First, we write down the objective function

$$\begin{aligned}
 \mathbb{E}_{\mathbf{Z} \sim \gamma_t(\mathbf{Z})} [\log p(\mathbf{X}, \mathbf{Z} | \mu_1, \dots, \mu_K)] &= \sum_{i=1}^N \sum_{k=1}^K \gamma_t(z_i = k) \log \left(\frac{1}{K} \prod_{d=1}^D \frac{(x_{id} \mu_{kd})^{(x_{id}-1)} \exp(-\mu_{kd} x_{id})}{x_{id}!} \right) \\
 &= C + \sum_{i=1}^N \sum_{k=1}^K \gamma_t(z_i = k) \sum_{d=1}^D ((x_{id} - 1) \log \mu_{kd} - \mu_{kd} x_{id}) \\
 &= C + \underbrace{\sum_{k=1}^K \sum_{d=1}^D \sum_{i=1}^N \gamma_t(z_i = k) ((x_{id} - 1) \log \mu_{kd} - \mu_{kd} x_{id})}_{=: \mathcal{L}_{kd}} \\
 &= C + \sum_{k=1}^K \sum_{d=1}^D \mathcal{L}_{kd}
 \end{aligned}$$

We see that the objective can be decomposed as a sum, where each parameter μ_{kd} is only encountered in a single term $\mathcal{L}_{kd}$.

Therefore, to find the optimal μ_{kd} , we only need to maximize the respective term $\mathcal{L}_{kd}$.

For this, we find the derivative of $\mathcal{L}_{kd}$ w.r.t. μ_{kd} and set it to zero:

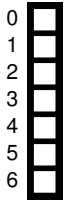
$$\begin{aligned}
 \frac{\partial \mathcal{L}_{kd}}{\partial \mu_{kd}} &= \frac{\partial}{\partial \mu_{kd}} \left(\sum_{i=1}^N \gamma_t(z_i = k) ((x_{id} - 1) \log \mu_{kd} - \mu_{kd} x_{id}) \right) \\
 &= \sum_{i=1}^N \gamma_t(z_i = k) \left(\frac{x_{id} - 1}{\mu_{kd}} - x_{id} \right) \stackrel{!}{=} 0.
 \end{aligned}$$

Setting the gradient to zero and solving for μ_{kd} , we obtain the update

$$\mu_{kd} = \frac{\sum_{i=1}^N \gamma_t(z_i = k) (x_{id} - 1)}{\sum_{i=1}^N \gamma_t(z_i = k) x_{id}}.$$

- 1 ✓ for correctly stating the objective $\mathbb{E}_{\mathbf{Z} \sim \gamma_t(\mathbf{Z})} [\log p(\mathbf{X}, \mathbf{Z} | \mu_1, \dots, \mu_K)]$ of the M-step
 - 5 ✓ for correctly simplifying the objective
 - 2 ✓ for the idea of computing the gradient w.r.t. μ_{kd} and setting it to zero
 - 2 ✓ for correctly computing the gradient
 - 2 ✓ for correctly solving for μ_{kd}
- Derive** keyword, i.e. 0 points for only stating the solution.

Problem 10 (Version A) (10 credits)



a)

We need to solve $\max_{\mathbf{x} \simeq \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_1$. ✓
Using the definition of f , this can be written as

$$\max_{\mathbf{x} \simeq \mathbf{x}'} \|\mathbf{A}(\mathbf{x} - \mathbf{x}')\|_1$$



Due to the definition of $\simeq$, $\mathbf{x}$ and $\mathbf{x}'$ can only differ in one component and at most by 1. Thus, the above is equivalent to

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \|\mathbf{A} \cdot (c \cdot \mathbf{e}_j)\|_1,$$

where $\mathbf{e}_j$ is the unit vector with non-zero entry in dimension j . ✓
The product with the unit vector is equivalent to summing over the j 'th column of $\mathbf{A}$, i.e.

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \left\| c \cdot \sum_{d=1}^4 A_{:,d} \right\|_1.$$



The function evidently has its maximum at $c = \pm 1$ and thus we have

$$\Delta_1 = \max_{j \in \{1,2,3,4\}} \left\| \sum_{d=1}^4 A_{:,d} \right\|_1.$$



To summarize: The maximum is achieved by changing dimension corresponding to the column whose sum has the largest absolute value. In our case, the Δ_1 -sensitivity is 24. ✓

Note: This is a "justify", not a "prove" task. If a student provides a reasonable / convincing argument for why the Δ_1 sensitivity corresponds to the absolute value of the largest column sum – for example by sketching the above proof in prose – they should also receive full credit.

Both for mathematical proofs and more intuitive explanations, make sure to give partial credits as indicated above (e.g. for using definition of Δ_1 sensitivity or plugging in the definition of f and $\simeq$).

Justify keyword, i.e. 0 points for only stating the value of Δ_1 .

b)

$\frac{24}{12}$ ✓

Correct solution is Δ_1/ϵ .

c)

This holds due to group-privacy ✓

$\mathbf{x} \simeq \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ up to 1 in exactly one dimension. $\mathbf{x} \simeq_\infty \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ by up to 1 in all four dimensions. Therefore, our mechanism is $4 \cdot \frac{1}{2} = 2$ -DP. ✓ ✓

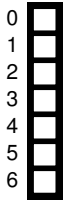
More formally: For any $\mathbf{x} \simeq_\infty \mathbf{x}'$ there is a sequence $\mathbf{x}_0, \dots, \mathbf{x}_4$ with $\mathbf{x}_0 = \mathbf{x}$, $\mathbf{x}_4 = \mathbf{x}'$ and $\mathbf{x}_j \simeq \mathbf{x}_{j+1}$. Because our mechanism is $\frac{1}{2}$ -DP w.r.t. $\simeq$, we have:

$$\begin{aligned} \mathbb{P}[\mathcal{M}_f(\mathbf{x}) \in Y] &= \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(0)}) \in Y] \\ &\leq e^{\frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(1)}) \in Y] \\ &\leq e^{2 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(2)}) \in Y] \\ &\dots \\ &\leq e^{4 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(t)}) \in Y] = e^2 \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}') \in Y], \end{aligned}$$

You can also give full credit if the students do not provide the last part of the proof but only explain why group-privacy applies (i.e. because $\mathbf{x} \simeq_\infty \mathbf{x}'$ means making 4 changes w.r.t $\simeq$).

Alternatively: Computing Δ_1 sensitivity w.r.t. " $\simeq_\infty$ ", like in subproblem a.) ✓ ✓ ✓

Problem 10 (Version B) (10 credits)



a)

We need to solve $\max_{\mathbf{x} \simeq \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_1$. ✓
Using the definition of f , this can be written as

$$\max_{\mathbf{x} \simeq \mathbf{x}'} \|\mathbf{A}(\mathbf{x} - \mathbf{x}')\|_1$$



Due to the definition of $\simeq$, $\mathbf{x}$ and $\mathbf{x}'$ can only differ in one component and at most by 1. Thus, the above is equivalent to

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \|\mathbf{A} \cdot (c \cdot \mathbf{e}_j)\|_1,$$

where $\mathbf{e}_j$ is the unit vector with non-zero entry in dimension j . ✓
The product with the unit vector is equivalent to summing over the j 'th column of $\mathbf{A}$, i.e.

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \left\| c \cdot \sum_{d=1}^4 A_{:,d} \right\|_1.$$



The function evidently has its maximum at $c = \pm 1$ and thus we have

$$\Delta_1 = \max_{j \in \{1,2,3,4\}} \left\| \sum_{d=1}^4 A_{:,d} \right\|_1.$$



To summarize: The maximum is achieved by changing dimension corresponding to the column whose sum has the largest absolute value. In our case, the Δ_1 -sensitivity is 20. ✓

Note: This is a "justify", not a "prove" task. If a student provides a reasonable / convincing argument for why the Δ_1 sensitivity corresponds to the absolute value of the largest column sum – for example by sketching the above proof in prose – they should also receive full credit.

Both for mathematical proofs and more intuitive explanations, make sure to give partial credits as indicated above (e.g. for using definition of Δ_1 sensitivity or plugging in the definition of f and $\simeq$).

Justify keyword, i.e. 0 points for only stating the value of Δ_1 .

b)

20 ✓

Correct solution is Δ_1/ϵ .

c)

This holds due to group-privacy ✓

$\mathbf{x} \simeq \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ up to 1 in exactly one dimension. $\mathbf{x} \simeq_\infty \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ by up to 1 in all four dimensions. Therefore, our mechanism is $4 \cdot \frac{1}{2} = 2$ -DP. ✓ ✓

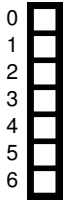
More formally: For any $\mathbf{x} \simeq_\infty \mathbf{x}'$ there is a sequence $\mathbf{x}_0, \dots, \mathbf{x}_4$ with $\mathbf{x}_0 = \mathbf{x}$, $\mathbf{x}_4 = \mathbf{x}'$ and $\mathbf{x}_j \simeq \mathbf{x}_{j+1}$. Because our mechanism is $\frac{1}{2}$ -DP w.r.t. $\simeq$, we have:

$$\begin{aligned} \mathbb{P}[\mathcal{M}_f(\mathbf{x}) \in Y] &= \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(0)}) \in Y] \\ &\leq e^{\frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(1)}) \in Y] \\ &\leq e^{2 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(2)}) \in Y] \\ &\dots \\ &\leq e^{4 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(t)}) \in Y] = e^2 \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}') \in Y], \end{aligned}$$

You can also give full credit if the students do not provide the last part of the proof but only explain why group-privacy applies (i.e. because $\mathbf{x} \simeq_\infty \mathbf{x}'$ means making 4 changes w.r.t $\simeq$).

Alternatively: Computing Δ_1 sensitivity w.r.t. " $\simeq_\infty$ ", like in subproblem a.) ✓ ✓ ✓

Problem 10 (Version C) (10 credits)



a)

We need to solve $\max_{\mathbf{x} \simeq \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_1$. ✓
Using the definition of f , this can be written as

$$\max_{\mathbf{x} \simeq \mathbf{x}'} \|\mathbf{A}(\mathbf{x} - \mathbf{x}')\|_1$$



Due to the definition of $\simeq$, $\mathbf{x}$ and $\mathbf{x}'$ can only differ in one component and at most by 1. Thus, the above is equivalent to

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \|\mathbf{A} \cdot (c \cdot \mathbf{e}_j)\|_1,$$

where $\mathbf{e}_j$ is the unit vector with non-zero entry in dimension j . ✓
The product with the unit vector is equivalent to summing over the j 'th column of $\mathbf{A}$, i.e.

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \left\| c \cdot \sum_{d=1}^4 A_{:,d} \right\|_1.$$



The function evidently has its maximum at $c = \pm 1$ and thus we have

$$\Delta_1 = \max_{j \in \{1,2,3,4\}} \left\| \sum_{d=1}^4 A_{:,d} \right\|_1.$$



To summarize: The maximum is achieved by changing dimension corresponding to the column whose sum has the largest absolute value. In our case, the Δ_1 -sensitivity is 18. ✓

Note: This is a "justify", not a "prove" task. If a student provides a reasonable / convincing argument for why the Δ_1 sensitivity corresponds to the absolute value of the largest column sum – for example by sketching the above proof in prose – they should also receive full credit.

Both for mathematical proofs and more intuitive explanations, make sure to give partial credits as indicated above (e.g. for using definition of Δ_1 sensitivity or plugging in the definition of f and $\simeq$).

Justify keyword, i.e. 0 points for only stating the value of Δ_1 .

b)

18 ✓

Correct solution is Δ_1/ϵ .

c)

This holds due to group-privacy ✓

$\mathbf{x} \simeq \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ up to 1 in exactly one dimension. $\mathbf{x} \simeq_{\infty} \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ by up to 1 in all four dimensions. Therefore, our mechanism is $4 \cdot \frac{1}{2} = 2$ -DP. ✓ ✓

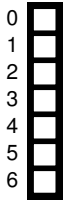
More formally: For any $\mathbf{x} \simeq_{\infty} \mathbf{x}'$ there is a sequence $\mathbf{x}_0, \dots, \mathbf{x}_4$ with $\mathbf{x}_0 = \mathbf{x}$, $\mathbf{x}_4 = \mathbf{x}'$ and $\mathbf{x}_j \simeq \mathbf{x}_{j+1}$. Because our mechanism is $\frac{1}{2}$ -DP w.r.t. $\simeq$, we have:

$$\begin{aligned} \mathbb{P}[\mathcal{M}_f(\mathbf{x}) \in Y] &= \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(0)}) \in Y] \\ &\leq e^{\frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(1)}) \in Y] \\ &\leq e^{2 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(2)}) \in Y] \\ &\dots \\ &\leq e^{4 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(t)}) \in Y] = e^2 \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}') \in Y], \end{aligned}$$

You can also give full credit if the students do not provide the last part of the proof but only explain why group-privacy applies (i.e. because $\mathbf{x} \simeq_{\infty} \mathbf{x}'$ means making 4 changes w.r.t $\simeq$).

Alternatively: Computing Δ_1 sensitivity w.r.t. " $\simeq_{\infty}$ ", like in subproblem a.) ✓ ✓ ✓

Problem 10 (Version D) (10 credits)



a)

We need to solve $\max_{\mathbf{x} \simeq \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_1$. ✓
Using the definition of f , this can be written as

$$\max_{\mathbf{x} \simeq \mathbf{x}'} \|\mathbf{A}(\mathbf{x} - \mathbf{x}')\|_1$$



Due to the definition of $\simeq$, $\mathbf{x}$ and $\mathbf{x}'$ can only differ in one component and at most by 1. Thus, the above is equivalent to

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \|\mathbf{A} \cdot (c \cdot \mathbf{e}_j)\|_1,$$

where $\mathbf{e}_j$ is the unit vector with non-zero entry in dimension j . ✓
The product with the unit vector is equivalent to summing over the j 'th column of $\mathbf{A}$, i.e.

$$\max_{j \in \{1,2,3,4\}} \max_{c \in [-1,1]} \left\| c \cdot \sum_{d=1}^4 A_{:,d} \right\|_1.$$



The function evidently has its maximum at $c = \pm 1$ and thus we have

$$\Delta_1 = \max_{j \in \{1,2,3,4\}} \left\| \sum_{d=1}^4 A_{:,d} \right\|_1.$$



To summarize: The maximum is achieved by changing dimension corresponding to the column whose sum has the largest absolute value. In our case, the Δ_1 -sensitivity is 32. ✓

Note: This is a "justify", not a "prove" task. If a student provides a reasonable / convincing argument for why the Δ_1 sensitivity corresponds to the absolute value of the largest column sum – for example by sketching the above proof in prose – they should also receive full credit.

Both for mathematical proofs and more intuitive explanations, make sure to give partial credits as indicated above (e.g. for using definition of Δ_1 sensitivity or plugging in the definition of f and $\simeq$).

Justify keyword, i.e. 0 points for only stating the value of Δ_1 .

b)

$\frac{32}{12}$ ✓

Correct solution is Δ_1/ϵ .

c)

This holds due to group-privacy ✓

$\mathbf{x} \simeq \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ up to 1 in exactly one dimension. $\mathbf{x} \simeq_\infty \mathbf{x}'$ means that $\mathbf{x}$ and $\mathbf{x}'$ differ by up to 1 in all four dimensions. Therefore, our mechanism is $4 \cdot \frac{1}{2} = 2$ -DP. ✓ ✓

More formally: For any $\mathbf{x} \simeq_\infty \mathbf{x}'$ there is a sequence $\mathbf{x}_0, \dots, \mathbf{x}_4$ with $\mathbf{x}_0 = \mathbf{x}$, $\mathbf{x}_4 = \mathbf{x}'$ and $\mathbf{x}_j \simeq \mathbf{x}_{j+1}$. Because our mechanism is $\frac{1}{2}$ -DP w.r.t. $\simeq$, we have:

$$\begin{aligned} \mathbb{P}[\mathcal{M}_f(\mathbf{x}) \in Y] &= \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(0)}) \in Y] \\ &\leq e^{\frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(1)}) \in Y] \\ &\leq e^{2 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(2)}) \in Y] \\ &\dots \\ &\leq e^{4 \cdot \frac{1}{2}} \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}^{(t)}) \in Y] = e^2 \cdot \mathbb{P}[\mathcal{M}_f(\mathbf{x}') \in Y], \end{aligned}$$

You can also give full credit if the students do not provide the last part of the proof but only explain why group-privacy applies (i.e. because $\mathbf{x} \simeq_\infty \mathbf{x}'$ means making 4 changes w.r.t $\simeq$).

Alternatively: Computing Δ_1 sensitivity w.r.t. " $\simeq_\infty$ ", like in subproblem a.) ✓ ✓ ✓

Problem 11 (Version A) (11 credits)

0
1

a)

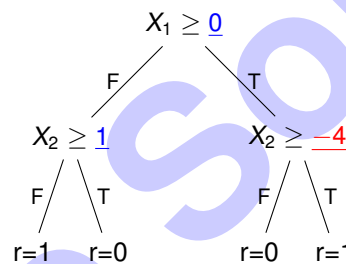
ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	1	0	0	1	1

Correctly computing all six predictions ✓

0
1
2
3

b)

Change -2 to -4 .



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	1	1	0	1	1

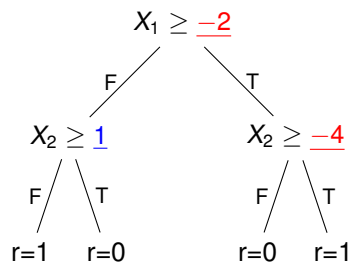
Modifying the tree such that $P(R = 1|A = a) = P(R = 1|A = b)$, i.e. both subpopulations have the same number of $R = 1$ ✓✓✓

If more than one decision threshold is changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

c)



Change -2 to -4 and 0 to -2 .



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	0	1	0	1	1

Modifying the tree such that $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$ and $P(R = 1|A = a, Y = 0) = P(R = 1|A = b, Y = 0)$, i.e. both subpopulations have the same True Positive Rate and False Positive Rate ✓✓✓

Alternatively: Modifying the tree such that we have 100% accuracy for both subpopulations ✓✓✓

If more than two decision thresholds are changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

0 ☐
1 ☐
2 ☐

d)

Yes, it exists. See subproblem b.).

Drawing any tree with $P(R = 1|A = a) = P(R = 1|A = b)$ and $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$, i.e. both subpopulations have the same number of $R = 1$ and the same False Positive rate

True Positive Rate and False Positive Rate ✓✓

Alternatively: Drawing a tree that always predicts 0 or always predicts 1 ✓✓✓

0 ☐
1 ☐
2 ☐

e)

No, because Y and A are correlated / not independent (see homework week 14, problem 3).

Referencing result from homework ✓✓ Alternatively: Repeating proof from homework (either verbatim or high-level summary) ✓✓

Justify keyword, i.e. 0 points for only stating the solution.

Problem 11 (Version B) (11 credits)

a)

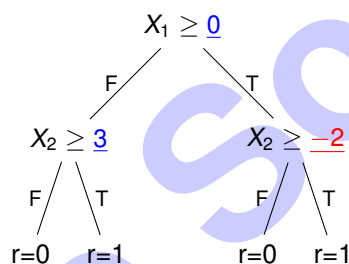
ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	1	0

Correctly computing all six predictions ✓

0
1

b)

Change 1 to -2 .



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	1	1

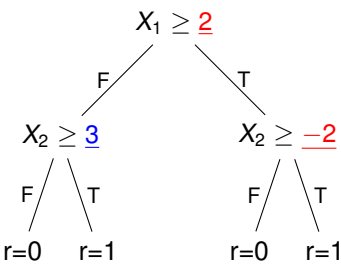
Modifying the tree such that $P(R = 1|A = a) = P(R = 1|A = b)$, i.e. both subpopulations have the same number of $R = 1$ ✓✓✓

If more than one decision threshold is changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

0
1
2
3

c)

Change 1 to -2 and 0 to 2.



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	0	1

Modifying the tree such that $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$ and $P(R = 1|A = a, Y = 0) = P(R = 1|A = b, Y = 0)$, i.e. both subpopulations have the same True Positive Rate and False Positive Rate ✓✓✓

Alternatively: Modifying the tree such that we have 100% accuracy for both subpopulations ✓✓✓

If more than two decision thresholds are changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

d)

Yes, it exists. See subproblem b.).

Drawing any tree with $P(R = 1|A = a) = P(R = 1|A = b)$ and $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$, i.e. both subpopulations have the same number of $R = 1$ and the same False Positive rate

True Positive Rate and False Positive Rate ✓✓

Alternatively: Drawing a tree that always predicts 0 or always predicts 1 ✓✓

e)

No, because Y and A are correlated / not independent (see homework week 14, problem 3).

Referencing result from homework ✓✓✓ Alternatively: Repeating proof from homework (either verbatim or high-level summary) ✓✓✓

Justify keyword, i.e. 0 points for only stating the solution.

Problem 11 (Version C) (11 credits)

0
1

a)

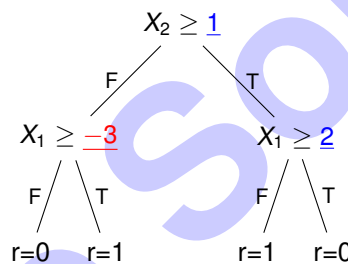
ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	1	0	0	1	1

Correctly computing all six predictions ✓

0
1
2
3

b)

Change -1 to -3 .



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	1	1	0	1	1

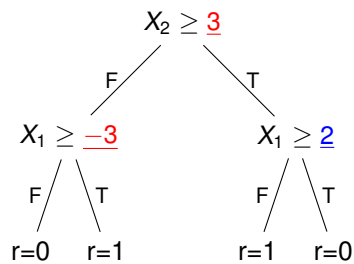
Modifying the tree such that $P(R = 1|A = a) = P(R = 1|A = b)$, i.e. both subpopulations have the same number of $R = 1$ ✓✓✓

If more than one decision threshold is changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

c)

	0
	1
	2
	3

Change -1 to -3 and 1 to 3 .



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	0	1	0	1	1
R	0	0	1	0	1	1

Modifying the tree such that $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$ and $P(R = 1|A = a, Y = 0) = P(R = 1|A = b, Y = 0)$, i.e. both subpopulations have the same True Positive Rate and False Positive Rate ✓✓✓

Alternatively: Modifying the tree such that we have 100% accuracy for both subpopulations ✓✓✓
 If more than two decision thresholds are changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

0 ☐
1 ☐
2 ☐

d)

Yes, it exists. See subproblem b.).

Drawing any tree with $P(R = 1|A = a) = P(R = 1|A = b)$ and $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$, i.e. both subpopulations have the same number of $R = 1$ and the same False Positive rate

True Positive Rate and False Positive Rate ✓✓

Alternatively: Drawing a tree that always predicts 0 or always predicts 1 ✓✓

0 ☐
1 ☐
2 ☐

e)

No, because Y and A are correlated / not independent (see homework week 14, problem 3).

Referencing result from homework ✓✓ Alternatively: Repeating proof from homework (either verbatim or high-level summary) ✓✓

Justify keyword, i.e. 0 points for only stating the solution.

Problem 11 (Version D) (11 credits)

a)

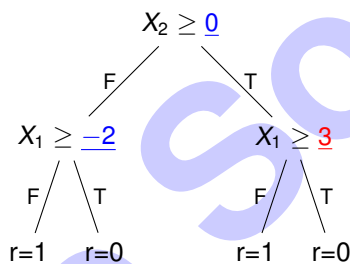
ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	1	0

Correctly computing all six predictions ✓

0
1

b)

Change child 0 to 3



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	1	1

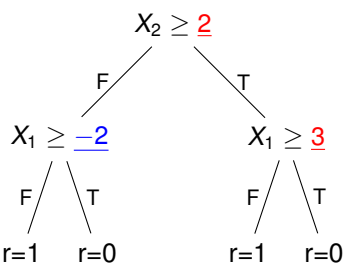
Modifying the tree such that $P(R = 1|A = a) = P(R = 1|A = b)$, i.e. both subpopulations have the same number of $R = 1$ ✓✓✓

If more than one decision threshold is changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

0
1
2
3

c)

Change child 0 to 3 and root 0 to 2.



Resulting predictions:

ID	1	2	3	4	5	6
A	a	a	a	b	b	b
Y	0	1	1	0	0	1
R	0	1	1	0	0	1

Modifying the tree such that $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$ and $P(R = 1|A = a, Y = 0) = P(R = 1|A = b, Y = 0)$, i.e. both subpopulations have the same True Positive Rate and False Positive Rate ✓✓✓

Alternatively: Modifying the tree such that we have 100% accuracy for both subpopulations ✓✓✓

If more than two decision thresholds are changed or the variables / inequality symbols are changed or nodes are added / removed: 0 points.

d)

Yes, it exists. See subproblem b.).

Drawing any tree with $P(R = 1|A = a) = P(R = 1|A = b)$ and $P(R = 1|A = a, Y = 1) = P(R = 1|A = b, Y = 1)$, i.e. both subpopulations have the same number of $R = 1$ and the same False Positive rate

True Positive Rate and False Positive Rate ✓✓

Alternatively: Drawing a tree that always predicts 0 or always predicts 1 ✓✓

e)

No, because Y and A are correlated / not independent (see homework week 14, problem 3).

Referencing result from homework ✓✓✓ Alternatively: Repeating proof from homework (either verbatim or high-level summary) ✓✓✓

Justify keyword, i.e. 0 points for only stating the solution.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

The image shows a large grid of graph paper, intended for writing solutions. A diagonal watermark is overlaid across the grid. The watermark consists of two parts: 'Sample Solution' in a large, light blue font, and 'Correction Notes' in a smaller, red font below it. The grid itself is composed of small squares, typical of graph paper used in mathematics or engineering.

Sample Solution

Correction Notes

Sample Solution

Correction Notes

Sample Solution
Correction Notes

Sample Solution

Correction Notes

Sample Solution
Correction Notes

Sample Solution

Correction Notes

Sample Solution
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Sample Solution

Correction Notes